LESSON 10.5

Math is Boundless

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

WHY THIS MATTERS Geometry is an integral part of design. Geometric patterns have been used to decorate homes and buildings for centuries — in art, architecture, and clothing. This lesson looks at three elements of geometric design: repetition, pattern, and rhythm.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
repetition *Spanish: repetición*	The same object, shape, or form repeated multiple times.	The same tile, over and over, all the way along the border.
pattern *Spanish: patrón*	A set of objects or shapes that are repeated and arranged in a regular order.	Circle, square, triangle, circle, square, triangle …
pattern unit *Spanish: unidad del patrón*	The smallest chunk of a pattern that repeats.	circle · square · triangle — the chunk that starts over.
rhythm *Spanish: ritmo*	A design element created by varying the placement, size, orientation, or order of the same objects or shapes.	The same circles, but sizes and spacing change as your eye travels.
predictability *Spanish: previsibilidad*	Being able to know what comes next because the design follows a rule.	Shape 25 must be a circle — the unit repeats every 3.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A border repeats: pink circle, purple square, blue triangle, pink circle, purple square, blue triangle, ...

1. The pattern unit is 3 shapes long: circle, square, triangle.
2. To predict shape number 12, I divide: 12 ÷ 3 = 4 with no remainder, so shape 12 ends a complete unit — it is the blue triangle.
3. The next three shapes after any complete unit will be a pink circle, a purple square, and a blue triangle.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The pattern unit is still 3 shapes long.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The pattern unit is _____, so this is _____ rather than _____.
- I have seen a geometric pattern on _____ .
- Its pattern unit is _____ .
- A design with only repetition feels _____ .

Word bank repetition | pattern | pattern unit | rhythm | predictability

WATCH OUT!

A common mistake

Calling every repeated design a 'pattern.' One identical shape repeated is repetition

LESSON 10.5

Math is Boundless

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

In geometric design, what is repetition?

(A) A set of different shapes arranged randomly
(B) Shapes that vary in size and order of appearance
(C) Any drawing that uses color
(D) The same object or shape repeated multiple times

HINT The word itself is a clue: something repeats.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A border shows: pink circle, purple square, blue triangle, pink circle, purple square, blue triangle. What are the next three shapes?

(A) Blue triangle, purple square, pink circle
(B) Pink circle, purple square, blue triangle
(C) Pink circle, blue triangle, purple square
(D) It is impossible to predict

HINT Find the chunk that repeats.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

In the circle-square-triangle border, how many shapes long is the pattern unit?

(A) 1 shape
(B) 2 shapes
(C) 3 shapes
(D) 6 shapes

HINT Where does the pattern start over?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is rhythm in geometric design?

(A) The same objects or shapes with varying placement, size, or order of appearance
(B) One shape repeated exactly, over and over
(C) Shapes arranged in a strict, predictable order
(D) A design with no shapes at all

HINT Rhythm uses the SAME shapes as a pattern might.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Where have you seen geometric patterns decorating a building, clothing, or artwork? Describe the design.

SENTENCE STARTER I have seen a geometric pattern on ___ .
Its pattern unit is ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

How would you compare the visual appeal of a design with just repetition and a design with patterns?

SENTENCE STARTER A design with only repetition feels ___ .
A design with patterns feels ___ because ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you called rhythm and your partner called repetition. What is the pattern unit, and what would have to change for it to stop being predictable?

SENTENCE STARTER The pattern unit is ___, so this is ___ rather than ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.5

Math is Boundless

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

The pattern unit circle-square-triangle repeats. What is shape number 12?

(A) The pink circle
(B) The blue triangle
(C) The purple square
(D) It cannot be determined

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Same pattern. What is shape number 20?

(A) The purple square
(B) The pink circle
(C) The blue triangle
(D) The 20th shape starts a new pattern

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

How many COMPLETE pattern units appear in the first 18 shapes of the circle-square-triangle border?

(A) 3 units

(B) 6 units

(C) 9 units

(D) 15 units

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How is a design with rhythm different from a pattern?

(A) Rhythm uses shapes and a pattern does not

(B) A pattern varies its shapes and rhythm never does

(C) There is no difference between them

(D) Rhythm varies size, orientation, and order, so it often lacks predictability; a pattern repeats in a regular order

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

A painting shows circles whose colors, sizes, and thickness vary, along with the number of concentric circles inside each one. Describe how the painting demonstrates rhythm.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

If a new row of circles were added to the rhythm painting, what would they look like? Explain your thinking.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you called rhythm and your partner called repetition. What is the pattern unit, and what would have to change for it to stop being predictable?

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LESSON 10.5

Math is Boundless

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each vertex arrangement to its angle sum — and what it means for tessellation.

_____ 4 squares at a vertex

_____ 6 equilateral triangles at a vertex

_____ 3 regular hexagons at a vertex

_____ 3 regular pentagons at a vertex

(A) 4 × 90° = 360° — no gaps

(B) 6 × 60° = 360° — no gaps

(C) 3 × 120° = 360° — no gaps

(D) 3 × 108° = 324° — a gap remains

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

The circle-square-triangle pattern continues. What is shape number 25?

(A) The purple square

(B) The blue triangle

(C) There is no shape 25

(D) The pink circle

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which element of geometric design often LACKS predictability?

(A) Rhythm
(B) Repetition
(C) Pattern
(D) The pattern unit

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Create a design that has repetition, pattern, AND rhythm. Describe (or sketch) it, and label where each element appears.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

Why do you think the lesson is called 'Math is Boundless'? Use repetition, pattern, and rhythm in your answer.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you called rhythm and your partner called repetition. What is the pattern unit, and what would have to change for it to stop being predictable?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help